

Understanding Dataset Difficulty with V-Usable Information

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- They note benchmarks can mislead despite steering progress.
- They note human-model gaps at instance level.
- Accuracy ignores confidence and predictive entropy.
- They seek a unitized, model-aware difficulty.
- They define difficulty as lack of V-usable info.
- They score instances with pointwise V-information.

Method: Predictive V-Information and PVI

- They estimate $I_V(X \rightarrow Y) = H_V(Y) - H_V(Y | X)$ by training null-input and full-input models in V .
- $PVI(x \rightarrow y) = -\log g[\emptyset](y) + \log g'[x](y)$; average PVI equals I_V .
- Negative PVI flags harmful or misleading inputs.
- Evaluation uses held-out data to estimate entropies reliably.

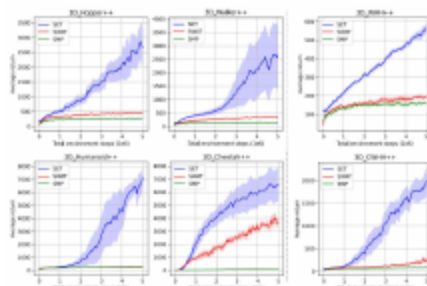


Figure: Histograms illustrate how PVI varies by outcome (correct vs. error).

Key Findings: What Makes Data Usable for V

- They find lexical identity (shuffled) retains most V-usable information; hypothesis-only remains informative; premise-only is near zero.
- They report SNLI > MNLI > CoLA in BERT-usable information.
- They use these analyses to reveal artefacts and slice-level difficulty.

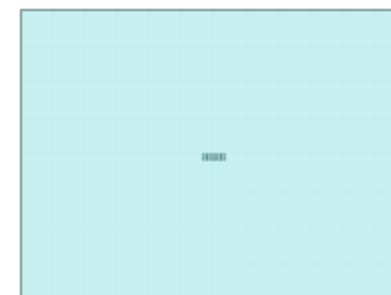


Figure: Attribute transformations on SNLI: shuffled (lexical identity) dominates, hypothesis-only informative, premise-only near zero.

- How stable are I_V and PVI under distribution shift or new domains?
- How should they extend I_V /PVI to open-ended generation and other modalities while preserving interpretability?
- Which transformations best isolate causal attribute contributions without inducing spurious signals?